

NANTES ATLANTIQUE AP, France

WMO: 072220

Lat: 47.1500N

Lon: 1.6089W

Elev: 90

StdP: 14.65

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.8	28.6	19.3	14.6	29.7	23.0	17.4	31.1	28.0	52.2	25.1	52.6	5.2	60	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	87.6	68.2	83.9	67.1	80.3	65.7	70.7	82.9	68.9	79.7	67.2	76.7	8.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	98.4	74.1	65.3	93.6	72.4	63.9	89.2	70.9	34.5	83.0	33.0	79.9	31.7	76.7	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.9	19.0	16.7	21.9	94.6	3.8	3.4	19.2	97.1	16.9	99.1	14.8	101.0	12.1	103.5		
			DB	WB	20.8	73.8	3.8	1.9	18.1	75.2	15.9	76.3	13.7	77.3	11.0	78.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.6	43.2	44.1	48.2	52.0	57.9	63.9	67.3	67.0	62.3	56.5	48.6	43.8
	DBStd	10.49	6.75	6.12	5.39	5.44	5.48	5.83	5.43	5.20	5.11	5.58	5.89	7.01
	HDD50	851	222	176	99	38	3	0	0	0	0	12	92	209
	HDD65	4091	677	586	522	390	230	89	36	33	112	268	492	656
	CDD50	2538	10	10	42	98	248	417	535	528	370	212	50	18
	CDD65	303	0	0	0	1	10	56	106	96	31	3	0	0
	CDH74	2692	0	0	0	16	110	535	934	816	265	16	0	0
	CDH80	830	0	0	0	0	15	160	309	282	63	1	0	0
Wind	WSAvg	8.0	9.1	8.9	8.8	8.5	7.9	7.6	7.4	7.0	7.1	7.4	7.8	8.7
Precipitation	PrecAvg	32.6	3.7	2.8	2.2	2.4	2.4	1.8	1.8	1.8	2.4	3.7	3.7	3.8
	PrecMax	43.2	7.0	5.9	5.9	6.9	5.4	4.9	4.7	4.0	8.7	8.7	7.8	6.7
	PrecMin	21.7	0.4	0.2	0.3	0.5	0.5	0.2	0.2	0.4	0.0	0.7	1.0	0.4
	PrecStd	5.0	1.9	1.5	1.4	1.5	1.1	1.0	0.9	1.1	2.0	1.7	1.9	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	61.3	69.2	76.4	82.4	90.9	92.8	93.3	86.5	76.0	64.1	58.4
		MCWB	53.9	52.9	56.2	61.0	65.1	70.1	69.4	70.3	66.4	64.6	58.6	55.3
	2%	DB	55.5	56.8	64.1	70.9	77.1	84.3	87.1	87.0	80.9	71.1	61.1	56.8
		MCWB	53.2	51.4	54.2	58.6	62.8	67.6	67.8	68.6	65.0	62.7	57.6	54.6
	5%	DB	54.0	54.2	60.2	66.1	73.1	80.0	83.3	82.6	76.4	67.5	59.2	55.4
		MCWB	52.0	50.7	52.5	55.5	61.8	65.8	66.6	67.1	63.5	61.1	56.4	53.4
	10%	DB	52.5	52.7	56.9	62.1	68.9	75.8	79.1	78.0	72.5	64.8	57.4	53.9
		MCWB	50.5	50.0	51.0	53.7	59.4	63.9	65.2	65.3	61.6	60.0	54.5	52.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.2	54.6	57.8	62.7	67.9	72.9	72.3	73.9	69.4	66.1	61.0	56.7
		MCDB	55.8	58.8	65.8	73.3	78.7	85.4	86.3	87.9	78.9	71.1	62.3	57.4
	2%	WB	53.9	53.0	55.4	59.4	64.7	69.5	69.9	70.8	67.1	64.0	58.5	55.2
		MCDB	54.9	55.3	61.3	68.4	74.1	81.2	82.9	82.6	76.3	68.6	60.3	56.3
	5%	WB	52.5	51.8	53.8	57.0	62.4	67.1	68.1	68.8	65.2	62.5	56.8	53.8
		MCDB	53.9	53.7	58.1	64.0	71.0	77.5	79.3	78.6	73.2	66.3	58.8	55.2
	10%	WB	50.9	50.5	52.3	54.7	60.2	64.9	66.6	66.8	63.4	60.9	55.0	52.3
		MCDB	52.4	52.6	55.9	60.5	67.4	73.9	76.3	74.7	70.2	64.0	57.2	53.8
Mean Daily Temperature Range	5% DB	MDBR	10.4	12.8	15.0	16.7	17.0	18.2	18.6	19.3	18.7	14.8	12.3	10.9
		MCDBR	9.8	14.3	20.1	22.6	23.1	24.4	25.0	26.1	24.4	17.2	12.6	10.1
		MCWBR	8.3	10.4	12.2	12.3	12.0	11.1	10.1	10.6	10.9	10.1	9.3	8.7
	5% WB	MCDBR	9.2	12.2	15.9	19.9	20.9	22.4	21.9	22.2	20.5	14.6	11.2	9.9
		MCWBR	8.3	9.8	10.5	11.7	11.6	11.3	10.0	10.2	10.7	9.5	9.2	9.2
Clear-Sky Solar Irradiance	taub	0.330	0.346	0.374	0.393	0.397	0.404	0.397	0.386	0.367	0.364	0.339	0.324	
	taud	2.427	2.384	2.304	2.254	2.291	2.307	2.307	2.328	2.376	2.408	2.419	2.429	2.406
	Ebn at Noon	238	258	267	272	275	273	273	272	267	248	232	224	
	Edh at Noon	22	28	35	40	40	40	39	35	31	27	22	20	
All-Sky Solar Radiation	RadAvg	351	639	1003	1462	1728	1925	1840	1630	1289	751	435	307	
	RadStd	33	69	110	155	131	170	99	132	91	60	41	36	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page